

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2013-14

Paper: GLB-201 — Elements of Mineralogy and Crystallography

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem II/16

*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory.*

SECTION — A

Question 1. What is magma? Explain the magmatic process of mineral formation. [14]

Question 2. Explain with neat diagrams, the determination of Miller indices in different crystal systems. [14]

Question 3. With a neat diagram, describe different parts and their functions in a petrological microscope. [14]

SECTION — B

Question 4. Answer the following parts: [14]

(a) Which among the following is a mineral? [7]

(i) Wood

(ii) Quartz

Give proper justification in support of your answer.

(b) What do you understand by the hardness of a mineral? Explain Mohs' scale of hardness. [7]

Question 5. Write the chemical composition and physical properties of **any four** of the following minerals underlining the diagnostic ones: $[4 \times 3\frac{1}{2} = 14]$

(a) Hematite

(b) Gypsum

(c) Calcite

(d) Kyanite

(e) Bauxite

Question 6. Write short notes on **any four** of the following: $[4 \times 3\frac{1}{2} = 14]$

- (a) Centre of symmetry
- (b) First Law of Crystallography
- (c) Birefringence
- (d) Pleochroism
- (e) Twinkling

SECTION — C

Question 7. Answer the following questions:

- (a) Whether the following statement is True or False — “On a streak plate the streak of topaz is yellow in color.” Justify your answer with reasons. [2]
- (b) Arrange the following minerals in decreasing order of resistance to weathering: [2]
Hornblende; K-feldspar; Olivine; Quartz.
- (c) Coal and petroleum are products of _____ process. (Fill in the blank) [1]
- (d) Differentiate between the terms ‘crystal’ and ‘crystalline’. [2]
- (e) Name the crystallographic axes of calcite. [1]
- (f) What is a prism face in a crystal? [1]
- (g) Write the crystal systems of calcite and albite. [2]
- (h) Draw the ray velocity surface for an anisotropic mineral. [1]
- (i) What is a parametral face of a crystal? [1]
- (j) Give one example of an isotropic mineral. [1]